MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION

Sidhant College of Engineering

A/P Sudumbre , Tal-Maval, Dist-Pune

Microproject

Academic Year 2025-26

Program Name:- Artificial Intelligence And Machine Learning **Scheme:-** K
Subject Code:- Microporossesor Programming (314321) **Semester:-** 4th

A microproject on

Generate Fibonacci series

Sr.No	Name	Enrollment No	Seat No
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2	Anjum Mirza	25213380062	278653
3	Dnyaneshwari Jambukar	24213380007	278627

Under the guidence of

Miss.V.D.Patel

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CERTIFICATE

This is to certify that **Miss. Sakshi Diwane,** Fourth Semester of Diploma in Artificial Intelligence And Machine Learning (AIML) Engineering of Institute, Siddhant college of Engineering (Code:1624) has completed the Self Learning Activity (SLA) satisfactorily in Course Microprocessor Programming (314321) for the academic year 2025-2026 as prescribed in the curriculum.

Place:-Sudumbare. Enrollment no:-24213380006

Date:- Seat no:-278626

Miss.V.D.Patel	**Mr.A.J.Gaikaad**	**Mr.M.N.Pohare**
Subject Teacher	H.O.D.	Diploma Co-ordinator

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This is to certify that **Miss. Anjum Mirza,** Fourth Semester of Diploma in Artificial Intelligence And Machine Learning (AIML) Engineering of Institute, Siddhant college of Engineering (Code:1624) has completed the Self Learning Activity (SLA) satisfactorily in Course Microprocessor Programming (314321) for the academic year 2025-2026 as prescribed in the curriculum.

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This is to certify that **Miss. Dnyaneshwari Jambukar.** Fourth Semester of Diploma in Artificial Intelligence And Machine Learning (AIML) Engineering of Institute, Siddhant college of Engineering (Code:1624) has completed the Self Learning Activity (SLA) satisfactorily in Course Microprocessor Programming (314321) for the academic year 2025-2026 as prescribed in the curriculum.

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Miss.V.D.Patel	**Mr.A.J.Gaikaad**	**Mr.M.N.Pohare**
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1. Introduction to Fibonacci Series :

The Fibonacci series is a sequence of numbers in mathematics where each number is the sum of the two previous numbers. The series usually begins with 0 and 1.

So the sequence becomes:-

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, …

This means:

0 + 1 = 1

1 + 1 = 2

1 + 2 = 3

2 + 3 = 5

The Fibonacci series was introduced by the Italian mathematician Leonardo Fibonacci in the 13th century.

The Fibonacci series is important in mathematics, computer programming, and science. It also appears in nature, such as in the arrangement of flower petals, shells, and leaves.

Therefore, the Fibonacci series helps us understand patterns in mathematics, nature, and computer algorithms.

2. History of Fibonacci Series :

The Fibonacci series became well known through the work of the Italian mathematician Leonardo Fibonacci. In the year 1202, he wrote a famous book called Liber Abaci, where he introduced this number sequence to Europe.

In this book, Fibonacci explained a rabbit population problem to show how the numbers grow. While solving this problem, he discovered the sequence:

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, …

However, the pattern had been known earlier in Indian mathematics before Fibonacci described it. His work helped spread the concept widely in European mathematics.

Today, the Fibonacci series is very important in mathematics, computer science, and natural pattern studies.

3. Explanation of Fibonacci Formula :

The Fibonacci formula explains how each number in the Fibonacci series is calculated.

In this series, every new number is obtained by adding the previous two numbers.

$F_n = F_{n-1} + F_{n-2}$

Initial Values

The Fibonacci series starts with two fixed numbers:

□

□

Step-by-Step Example :-

Term (n)	conclusion	result
0	Given	0
1	Given	1
2	0+1	1
3	1+1	2
4	1+2	3
5	2+3	5
6	3+5	8

Simple Explanation

Start with 0 and 1.

Add the previous two numbers to get the next number.

Repeat the same process to continue the series.

So, the Fibonacci formula helps us generate the sequence step by step using the previous two numbers.

4. Algorithm For Fibonacci Series :

Step 1:- Start.

Step 2:- Input the number of terms n.

Step 3:- Initialize two variables.

a = 0

b = 1

Step 4:- Display a and b.

Step 5:- Repeat the following steps from i = 3 to n

c = a + b

Display c

a = b

b = c

Step 6:- Stop.

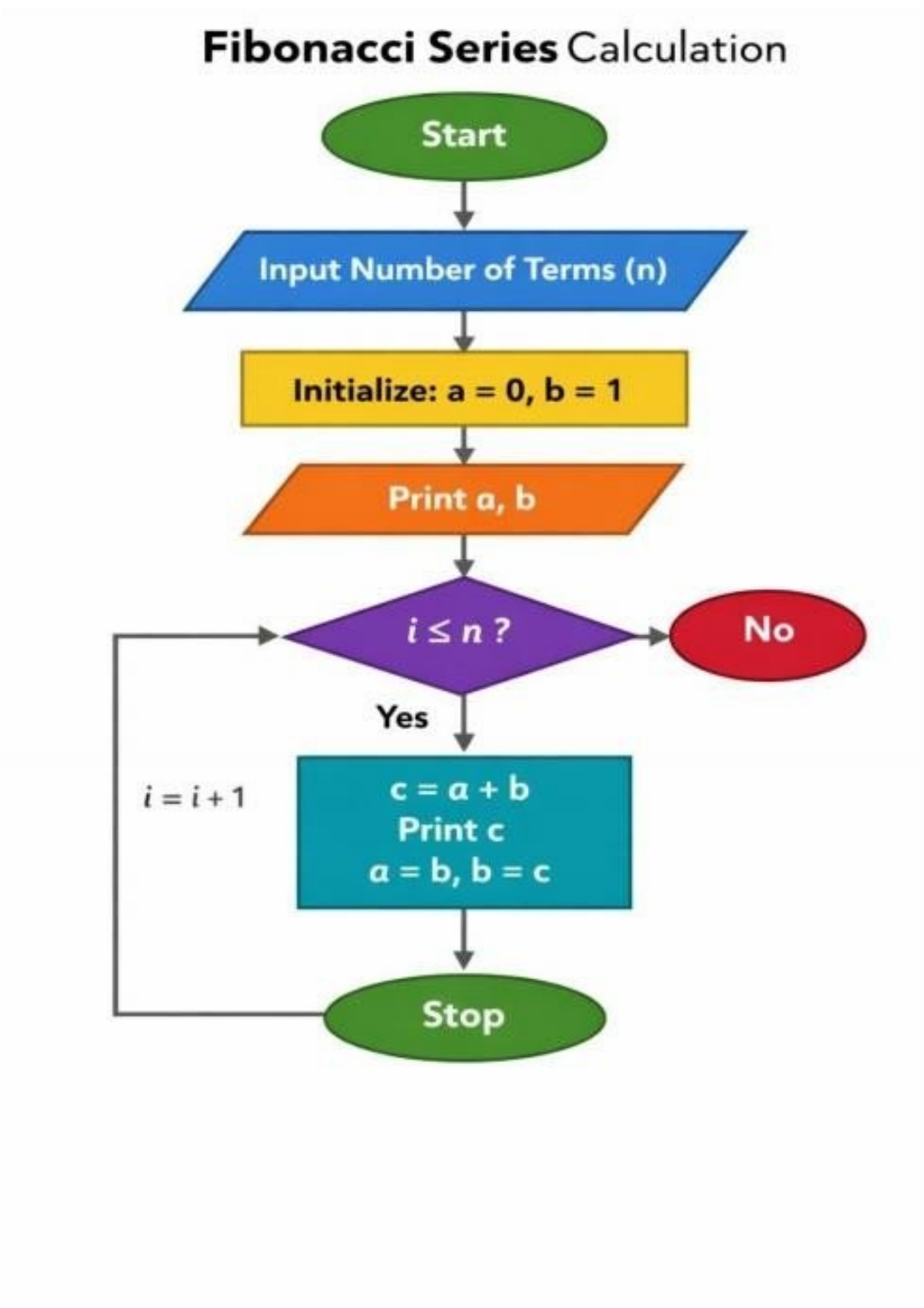
Simple Explanation

Start the series with 0 and 1

Add the previous two numbers to get the next number

Continue the process until n terms are generated.

5.Flowchart of Fibonacci series



6 Example Output of Program :

When the program runs, the user enters the number of terms. The program then prints the Fibonacci series.

Example:

Input:

Enter number of terms: 10

Output:

Fibonacci Series:

0 1 1 2 3 5 8 13 21 34

Explanation:

The program starts with 0 and 1.

Each next number is found by adding the previous two numbers.

The program continues until 10 terms are printed.

7. Fibonacci Series in Mathematics :

The Fibonacci series plays an important role in mathematics. It helps in understanding number patterns, sequences, and mathematical relationships.

In mathematics, the Fibonacci series is used to study recursion, series, and mathematical models. Many mathematical problems and algorithms are based on the Fibonacci sequence.

The series also helps in understanding the golden ratio, which appears when the ratio of two consecutive Fibonacci numbers approaches a constant value.

Mathematicians use the Fibonacci series in number theory, algebra, and discrete mathematics to analyze patterns and solve problems.

Therefore, the Fibonacci series is an important concept that helps in studying sequences, patterns, and mathematical structures.

8. Fibonacci Series in Nature :

The Fibonacci series can be observed in many natural patterns. Many plants and living things follow this pattern in their growth and structure.

For example, the number of petals in many flowers often follows Fibonacci numbers such as 3, 5, 8, 13, or 21. The arrangement of leaves around a stem also follows the Fibonacci pattern to help plants get maximum sunlight.

The spiral patterns seen in pinecones, sunflowers, and seashells also follow Fibonacci numbers. These patterns help nature grow in an efficient and balanced way.

Therefore, the Fibonacci series shows how mathematical patterns exist in nature, making it an important concept for studying natural designs and growth patterns.

9. Fibonacci Series in Computer Science :

The Fibonacci series is widely used in computer science to understand programming concepts and algorithms.

In computer science, Fibonacci numbers are used to learn recursion, loops, and algorithm design. Many programming languages like Python, Java, and C use Fibonacci programs to teach basic programming logic.

The Fibonacci sequence is also used in data structures and algorithms, such as dynamic programming and Fibonacci search technique. These methods help improve the efficiency of computer programs.

Therefore, the Fibonacci series is important in computer science because it helps students learn programming logic, algorithm development, and problem-solving techniques.

10. Applications of Fibonacci Series :

The Fibonacci series has many applications in different fields such as mathematics, computer science, nature, and finance.

1. Computer Programming :-

It is used to learn programming concepts, recursion, and algorithms.

2. Mathematics :-

The Fibonacci series helps in studying number patterns, sequences, and mathematical relationships.

3. Nature :-

Fibonacci numbers appear in flowers, pinecones, sunflowers, and shells.

4. Computer Algorithms :-

It is used in searching and sorting algorithms to improve program efficiency.

5. Financial Analysis :-

In the stock market, Fibonacci numbers are used to study price patterns and trends.

6. Art and Architecture :-

Fibonacci patterns are used to create balanced and attractive designs.

11. Advantages of Fibonacci Series :

The Fibonacci series has several advantages in mathematics and computer science.

1.Easy to Understand :-

The Fibonacci series follows a simple rule: each number is the sum of the two previous numbers, so it is easy to learn.

2. Helps in Learning Programming :-

It is widely used to teach loops, recursion, and algorithm design in programming.

3. Shows Mathematical Patterns :-

The series helps in understanding number patterns and sequences in mathematics.

4. Useful in Algorithms :-

Fibonacci numbers are used in searching techniques and algorithm optimization.

5. Appears in Nature :-

The Fibonacci pattern can be seen in flowers, plants, shells, and other natural structures.

12. Conclusion :

The Fibonacci series is an important mathematical sequence where each number
Input:

Enter number of terms: 10

Output:

Fibonacci Series:

0 1 1 2 3 5 8 13 21 34

Explanation:

The program starts with 0 and 1.

Each next number is found by adding the previous two numbers.

The program continues until 10 terms are printed.

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